SIMATS ENGINEERING

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# TECH STAR SUMMIT 2024

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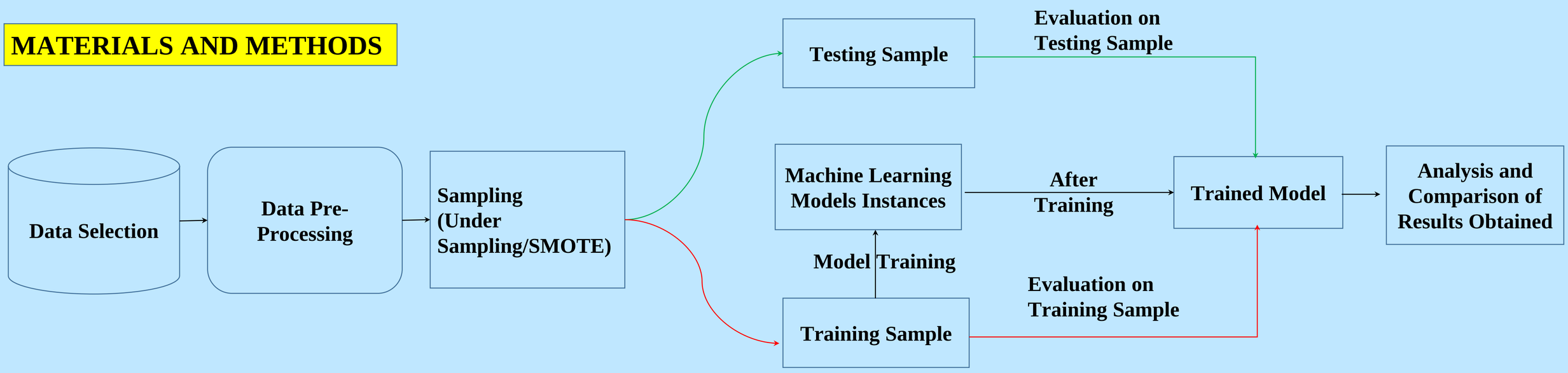
# Fraudulent Scam Call Detection using KNN Algorithm Compared with SVM-RBF Kernel Algorithm with Improved Accuracy

## INTRODUCTION

- The objective of this project is to enhance the accuracy of a fraudulent scam call detection , by comparing the performance of KNN algorithm with a SVM-RBF kernel algorithm.
- Scam calls may seek to extract sensitive data, including personally identifiable information (PII). Scam calls can also target businesses, attempting to trick employees or compromise sensitive corporate information. Fraudulent scam call detection initiatives contribute to public awareness and education.
- The application of fraudulent scam call detection are Telecommunications, Financial Institutions, Ecommerce Platforms, Government Agencies, Health Care and Insurance Companies .
- The advantage of using KNN algorithm has given better accuracy compared with SVM-RBF kernel algorithm.
- The dataset used was the CDR-Call-Details dataset obtained from Kaggle, consisting of 25,000 call details records and with a size of 2 MB. The dataset was divided into a training set of 19,330 details and a test set of 5,670 details.

Fig 1. Detect and Block Scam Calls

## MATERIALS AND METHODS




Fraudulent Scam Call Detection Flow Diagram

## RESULTS

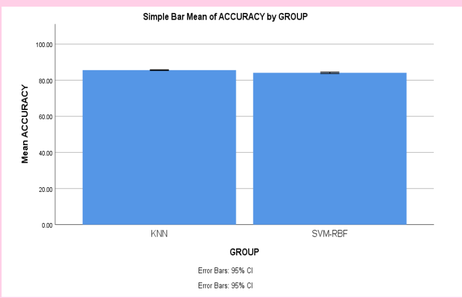



Fig 2. Accuracy value comparison of KNN and SVM-RBF Kernel Algorithm.

Table 1. Mean, Standard Deviation, Standard Error Mean with an accuracy rate comparison of KNN over SVM-RBF Kernel Algorithm .

| Group Statistics | | | | |
|---|---|---|---|---|
| Group Name | N | Mean | Standard Deviation | Std. Error Mean |
| KNN | 20 | 85.57 | 0.399 | 0.089 |
| SVM-RBF Kernel | 20 | 84.09 | 0.806 | 0.180 |

Table 2. A Significant Threshold value of an Accuracy rate comparison of KNN and SVM-RBF Kernel Algorithm.

| Independent Sample Test | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Levene's Test for Equality of Variances | | | T-test for Equality of Means | | | | | | |
| | F | Sig. | T | Df | Sig. (2-Tailed) | Mean Difference | Std. Error Differences | 95% Confidence Interval of the Difference | |
| | | | | | | | | Lower | Upper |
| Equal Variances assumed | 2.59 | .115 | 7.32 | 38 | .000 | 1.47 | .20 | 1.06 | 1.88 |
| Equal Variances not assumed | | | 7.32 | 27.7 | .000 | 1.47 | .20 | 1.06 | 1.88 |

In the present work, K Nearest Neighbor (KNN) is compared with SVM-RBF Kernel (SVM-RBF) , it depicts that the proposed algorithm gives more accuracy then compared algorithm.

## DISCUSSION AND CONCLUSION

- Based on T-test Statistical analysis, the significance value of p=0.000 (independent sample T - test p<0.05) is obtained and shows that there is a statistical significant difference between the group 1 and group 2.
- In comparison of accuracy, K Nearest Neighbor (KNN ) achieved 85.57% , outperforming the SVM-RBF Kernel (SVM-RBF) which achieved 84.09%, with an independent sample test indicating a significant difference in accuracy between the (p<0.05), with KNN achieving notably higher accuracy.
- The future scope for fraudulent scam call detection is likely to involve the integration of advanced technologies and the development of more sophisticated methods to combat evolving fraud techniques. The limitation of the research work is that it is supported for larger datasets, which is a restriction of the proposed work.
- The study presents fraudulent scam call detection and compares the accuracy of algorithms. The findings suggest that KNN algorithm (86%) is a better choice for scam call detection compared to SVM-RBF Kernel algorithm (85%).

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